
Supplementary information

Multifunctional ytterbium oxide buffer for perovskite solar cells

In the format provided by the
authors and unedited

Supporting Information

Multifunctional ytterbium oxide buffer for perovskite solar cells

Authors: Peng Chen^{1,11}, Yun Xiao^{2,11}, Juntao Hu^{3,10,11}, Shunde Li^{1,11}, Deying Luo^{4*}, Rui Su¹, Pietro Caprioglio², Pascal Kaienburg², Xiaohan Jia¹, Nan Chen³, Jingjing Wu⁵, Yanping Sui⁵, Pengyi Tang⁵, Haoming Yan¹, Tianyu Huang¹, Maotao Yu¹, Qiuyang Li¹, Lichen Zhao¹, Cheng-Hung Hou⁶, Yun-Wen You⁶, Jing-Jong Shyue⁶, Dengke Wang³, Xiaojun Li⁷, Qing Zhao¹, Qihuang Gong^{1,8,9*}, Zheng-Hong Lu^{3,4*}, Henry J. Snaith^{2*}, Rui Zhu^{1,8,9*}

¹State Key Laboratory for Artificial Microstructure and Mesoscopic Physics, School of Physics, Frontiers Science Center for Nano-optoelectronics & Collaborative Innovation Center of Quantum Matter, Peking University, 100871 Beijing, China

²Clarendon Laboratory, Department of Physics, University of Oxford, Oxford OX1 3PU, UK

³Department of Physics, Center for Optoelectronics Engineering Research, Yunnan University, 650091 Kunming, China

⁴Department of Materials Science and Engineering, University of Toronto, Toronto, Ontario M5G 3E4, Canada

⁵State Key Laboratory of Information Functional Materials, 2020 X-Lab, Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, Shanghai 200050, China

⁶Research Center for Applied Sciences, Academia Sinica, Taipei 11529, Taiwan

⁷Beijing National Laboratory for Molecular Sciences, CAS Key Laboratory of Organic Solids, Institute of Chemistry, Chinese Academy of Sciences, Beijing, 100190 China

⁸Peking University Yangtze Delta Institute of Optoelectronics, Nantong, 226010 Jiangsu, China

⁹Collaborative Innovation Center of Extreme Optics, Shanxi University, Taiyuan, 030006 Shanxi, China

¹⁰Present address: Department of Physics, Mathematics and Computer Science, Faculty of Basic Medical Science, Kunming Medical University, Kunming, 650500, Yunnan, China.

¹¹These authors contributed equally to this work

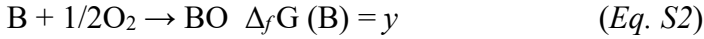
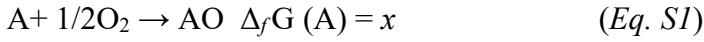
*e-mail: deying.luo@utoronto.ca (D.L.); qhgong@pku.edu.cn (Q.G.);

zhenghong.lu@utoronto.ca (Z.-H.L.); henry.snaith@physics.ox.ac.uk (H.J.S.);

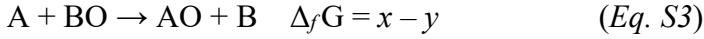
iamzhurui@pku.edu.cn (R.Z.)

Supplementary Note 1. Introduction of Gibbs free energy

Gibbs free energy is an essential parameter in assessing chemical thermodynamics, defined by $\Delta_f G = H - TS$, where H is the enthalpy, S is the entropy, and T is the thermodynamic temperature. It is well known that a reaction is thermodynamically favored if $\Delta_f G$ is negative. The $\Delta_f G$ of an oxide formation can also predict whether an oxide is stable if it directly contacts a metal. For example, consider two metals, A and B, and their oxidation reaction:



Using Equation (S2) minus Equation (S1), we get Eq. S3:



Here, the $\Delta_f G$ is the overall Gibbs energy of Eq. S3

If $x < y$, metal A will spontaneously reduce oxide BO. On the contrary, the reaction will not take place.

Compared to other common metal oxides used to modify interfaces in p-i-n PSCs, YbO_x presents the lowest standard molar Gibbs free energy of formation (**Extended Data Fig. 1b**)¹, indicating that the metal Yb tends to form YbO_x *via* chemical reaction with residual oxygen in a typical vacuum chamber when fabricated using a scalable thermal evaporation approach (**Extended Data Fig. 1c**)². Notably, such benefits are desirable for

making chemically stable buffer layers through a cost-effective, scalable thermal evaporation approach.

Supplementary Note 2. The industrial-scale application strengths of the YbO_x

Ytterbium (Yb), a key component for YbO_x, is a common rare-earth metal with a face-centered cubic crystal structure. Yb has a large atomic radius of 2.4 Å and can be processed through thermal evaporation in a vacuum chamber due to its low evaporation temperature of 736 K with a vapor pressure of 1 Pa [e.g., Cu (1,509 K), Ag (1,283 K), Au (1,646 K)]. We further compare the commercial applicability of YbO_x, SnO₂, and BCP (**Supplementary Table 1**). Due to the low usage of Yb for the production of large-scale YbO_x-based devices, the YbO_x shows the great cost advantages for installing a 1000-GW PV system. In addition, it takes only a few minutes to deposit the YbO_x layer by a simple thermal evaporation method, which is much less time-consuming compared to the fabrication of SnO₂ and BCP especially for the SnO₂ mostly fabricated by the temporal atomic layer deposition technique. To our knowledge, the fabrication of SnO₂ by widely-used temporal ALD consists of 4 processes for each cycle of material growth, and each cycle takes tens of seconds. For a typical temporal ALD-deposited SnO₂ with a thickness of 20 nm, it takes around several hours for the whole deposition process, which is highly time-consuming. The production duration for ALD SnO₂ will go down significantly when replacing the classic temporal ALD instrument with a high-throughout spatial ALD instrument. However, the cost for the spatial ALD instrument will be increased significantly (e.g., the quotation from the company is up to 3-fold higher than that of the current classic ALD instrument)³⁻⁵.

Furthermore, the reserves of these buffer materials in the long term must be considered . Even though Yb is regarded as a kind of rear-earth metal, the reserve of this metal is not as rare as its name might imply. Due to Yb's low usage and high reserve, YbO_x is predicated to guarantee the stable production of perovskite photovoltaics. Furthermore, our price comparison of metal Yb, and other common metals like Au, Ag, and Cu is listed in **Supplementary Table 2**. It is clear that the raw material cost for the fabrication of YbO_x is much lower than that of Au and Ag. To sum up, YbO_x shows three key advantages for commercialization: 1) low cost, 2) facile processing, and 3) vast raw material reserves.

Supplementary Note 3. The detailed oxidization process of the metal Yb

Many metals show oxidative behavior when exposed to oxygen. At the very beginning, oxidation is very rapid. However, after a few minutes or hours, this process slows down quickly, even becoming negligible, resulting in the formation of several nanometer-thick oxide layers at the surface. Regarding the Yb, such an oxidation process would be easy due to its low Gibbs free energy. After depositing a 1.5-nm-thick film by thermal evaporation in a vacuum chamber with a pressure of 4×10^{-4} Pa, in-situ XPS measurements detected a purely Yb^{3+} pattern without any Yb^0 signal as shown in **Extended data Figs. 2, 4a**. To quantify the rate of Yb oxidation, we recorded the oxidation behavior of a piece of clean Yb bulk prepared by 2400 s of Ar ion sputtering in the UHV (10^{-8} Pa) XPS chamber. **Extended data Fig. 4b** records the evolution of the Yb 4d spectra as exposure dose increased, and the $\text{Yb}^3\text{:Yb}^0$ ratios were determined using a nonlinear least squares fitting; then, the thickness of the oxidized film was calculated by the Lambert-Bier law (**Extended data Fig. 4c**). **Extended data Fig. 4d-i** displays the details of the fitting curves. The thickness of the oxidized layer shows an exponential increase with the exposed dose. When the clean Yb^0 surface is exposed to 480 Langmuir (**Supplementary Table 3**), the resulting thickness of the oxidized layer is about 1.5 nm. Therefore, it can be extrapolated that the 1.5-nm-thick Yb film has been completely oxidized in a few minutes with a background pressure of 4×10^{-4} Pa.

Supplementary Note 4. Charge transport in the YbO_x films through hopping

It is known that amorphous oxide materials contain various defects such as dangling bonds, oxygen vacancies, *etc.* These defects produce localized states within the band gap. If the density of such localized “gap states” is high enough, with energy levels near the Fermi level, electrons can undergo hopping transport from one site to another. This hopping transport theory was proposed by Mott and has been regularly adopted for modelling charge transport in organic semiconductors and disordered metal oxides⁴. In our case, we find that our prepared YbO_x films are amorphous when deposited on organic semiconductor underlying layers. Apart from the non-crystalline nature of the YbO_x, the valence band spectra explicitly show that the YbO_x films monitored by XPS shown in **Extended data Fig. 7a** have a high density of gap states below the Fermi level (**Extended data Fig. 7b**). To investigate the charge carrier transport in the YbO_x film, we fabricated the samples with the configuration of ITO/[YbO_x (2 nm)]₂₀/Cu. **Extended data Fig. 7c, d** reveals hopping charge-carrier transport behaviors followed by Mott’s model inside the YbO_x films. To further demonstrate the charge-carrier transport within the realistic PSCs, we fabricated the samples with a configuration of ITO/[YbO_x (5 nm)/Cu (1 nm)]₂₀/Cu. We conducted the temperature-dependent $J-V$ characteristics test (**Extended data Fig. 8a**). To complete the device fabrication, bilayers consisting of YbO_x (5 nm)/Cu (1 nm) were alternatively deposited (20 times). The conductivity extracted

against temperature shows trends in line with Mott's model⁶ (**Extended data Fig. 8b**), with the exponential power of $-1/T^{1/4}$.

Supplementary Note 5. Quasi Fermi-level Splitting (QFLS) calculation from EQE_{EL}

The following approach follows the report by Rau et al⁷. Briefly, the black body photon flux is

$$\phi_{BB} = \frac{1}{4\pi^2 \hbar^3 c^2} \frac{E^2}{\exp\left(\frac{E}{k_B T}\right) - 1} \quad (Eq. S4)$$

with $\hbar$ Plank's constant, k_B Boltzmann constant and T temperature. Assuming that the perovskite solar cell is at 300 K in thermal equilibrium with its environment, the dark radiative recombination current is:

$$J_{em,0} = q \int EQE_{PV}(E) \phi_{BB}(E) dE = J_{rad,0} \quad (Eq. S5)$$

with EQE_{PV} the photovoltaic *external* quantum efficiency of the perovskite solar cell and $J_{em,0}$ the current giving rise to emission, which also defines the dark radiative recombination current at $V = 0$. In this case, $J_{rad,0}$ has been calculated using the Shockley-Queisser approach for each different bandgap⁸. From that, the radiative limit of the QFLS ($QFLS_{rad}$) ($EQE_{EL} = 1$) can be calculated from the following equation:

$$QFLS_{rad} = \frac{k_B T}{q} \ln \left(\frac{J_G}{J_{rad,0}} \cdot 1 \right) \quad (Eq. S6)$$

where J_G is the generation current under illumination, in this case, approximated to the short circuit current J_{sc} of each bandgap.

We calculated the radiative V_{OC} of our perovskite absorber using Eq. S6.

$$QFLS_{rad\ WG} = \frac{k_B T}{q} \ln \left(\frac{J_{sc}}{J_{0,rad}} \right) = 1.494\ V \quad (Eq. S7)$$

$$QFLS_{\text{rad NG}} = \frac{k_B T}{q} \ln \left(\frac{J_{SC}}{J_{0,\text{rad}}} \right) = 1.279 \text{ V}$$

Moreover, by using the EQE_{EL} at the same injection condition as 1-sun equivalent measured for each different bandgap, it is possible to calculate the device QFLS, hence V_{OC} , with respect to the radiative QFLS potential

$$QFLS_{\text{WG}} = QFLS_{\text{rad WG}} + \frac{k_B T}{q} \ln (EQE_{\text{EL, WG}}) = 1.298 \text{ eV} \quad (\text{Eq. S8})$$

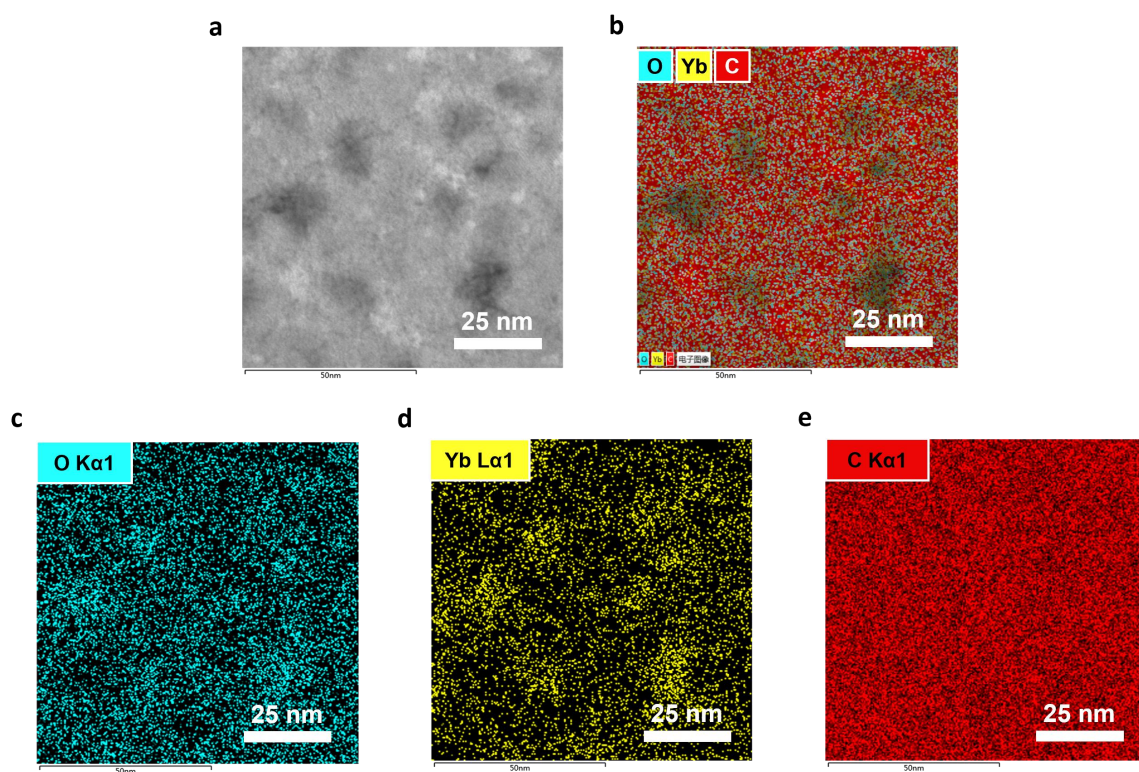
$$QFLS_{\text{NG}} = QFLS_{\text{rad NG}} + \frac{k_B T}{q} \ln (EQE_{\text{EL, NG}}) = 1.159 \text{ eV}$$

Supplementary Note 6. The suppression of bi-directional harmful species movements by inserting YbO_x

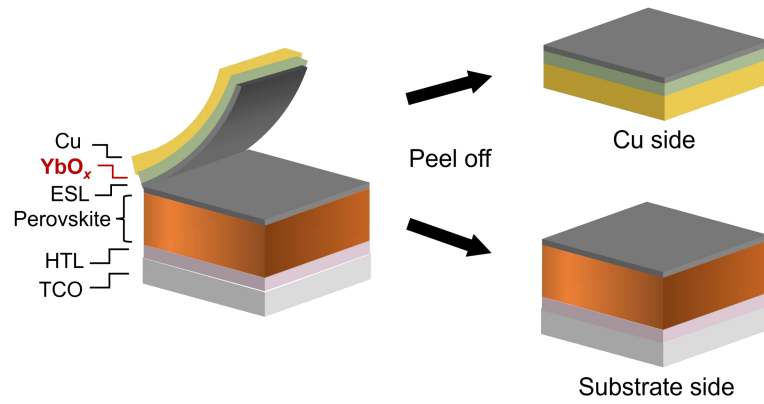
We note that using the YbO_x buffer layer in p-i-n PSCs effectively suppresses the diffusion of the evaporated Cu into the underlying C₆₀ layer and ion migration (such as I⁻) before accelerated ageing (**Extended Data Fig. 9**). With accelerated ageing, the bi-directional movement of atoms and ions for the control device with the BCP buffer layer have been significantly enhanced. Specifically, the evaporated metal Cu considerably diffuses into the perovskite absorber, while cations (such as Cs⁺ and Rb⁺) from the perovskite absorber migrate to the metal Cu electrode interface. Notably, these bi-directional movements of harmful species can be inhibited simply by replacing BCP with YbO_x (**Extended Data Fig. 9**). These observations agree with the depth-profile XPS results (**Fig. 2** and **Supplementary Fig. 18**). The consistent variation of the bi-directional species movements throughout the full PSCs before and after ageing suggest that YbO_x as an effective buffer layer can prevent close contact between the Cu electrode and the perovskite absorber, which has been shown to be helpful in maintaining the integrity of the metal Cu electrode under accelerated ageing conditions (**Supplementary Fig. 4**).

Additionally, the TOF-SIMS depth profiles reveals that a certain amount of Yb may present on the perovskite surface regardless of the accelerated ageing. It is thus urgently needed to investigate whether the YbO_x buffer layer used in the PSCs can decompose the perovskite surface. Therefore, we examined time-dependent XPS spectra of the

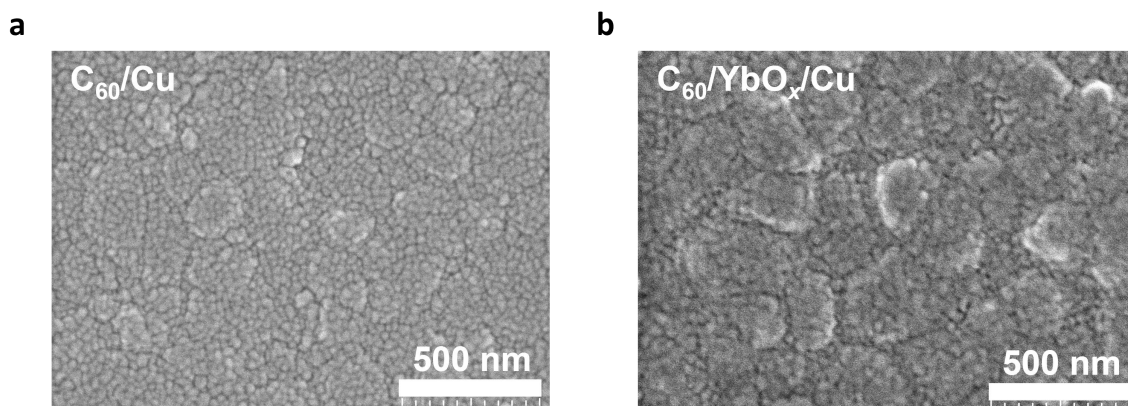
perovskite in direct contact with YbO_x (referred to as “target”). The isolated perovskite sample without YbO_x (referred to as “control”) is used for comparison. As shown in **Fig. 4e, f**, we find that after 25 days, both Pb 4f and I 3d XPS spectra of both the control and target samples show consistent characteristic peaks assigned to I^- and Pb^{2+} . And the Yb 4d XPS spectra show negligible changes over time (**Fig. 4g**). Note that the characteristic peak at 181 eV is ascribed to the Br 3p_{3/2}. These results suggest that the YbO_x is chemically unreactive with the perovskites. A partial YbO_x on the perovskite surfaces does not deteriorate the inherent material properties, thereby contributing to excellent device stability. We thus perceive that implementing YbO_x as the buffer material will help accelerate the rate of commercial application of perovskite photovoltaics through concurrent enhancements of both device efficiency and stability



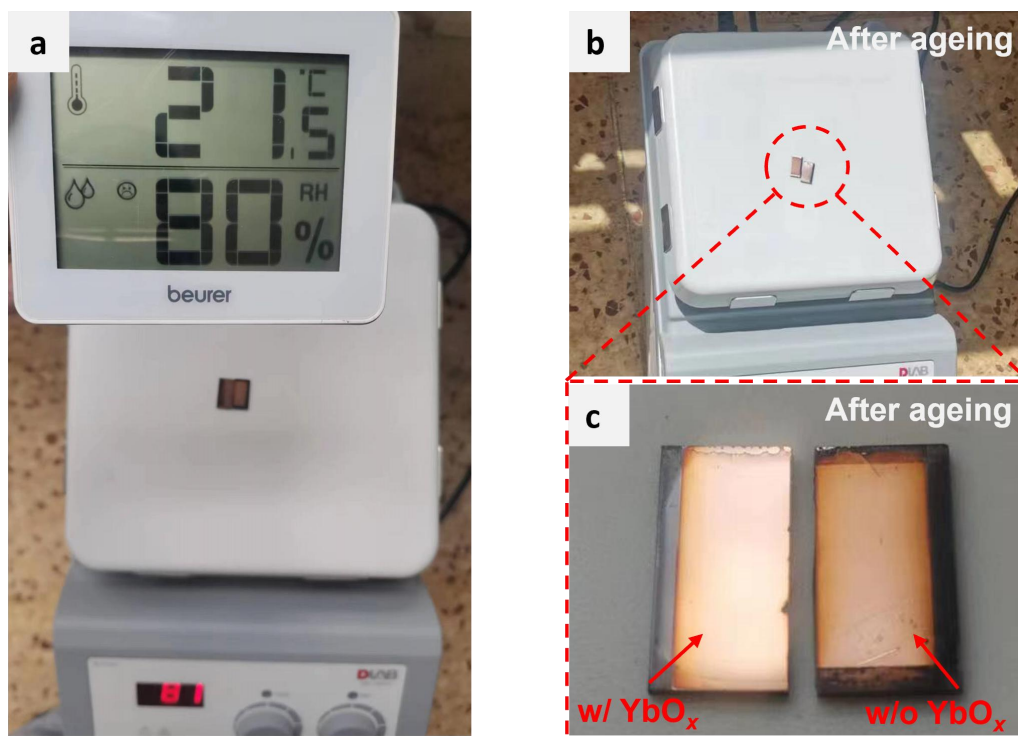
Supplementary Fig. 1. **a**, Representative high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images of the YbO_x deposited on the C_{60} -coated carbon-supported copper grid. **b-e**, Corresponding energy disperse X-ray (EDX) chemical composition maps in a selected region.



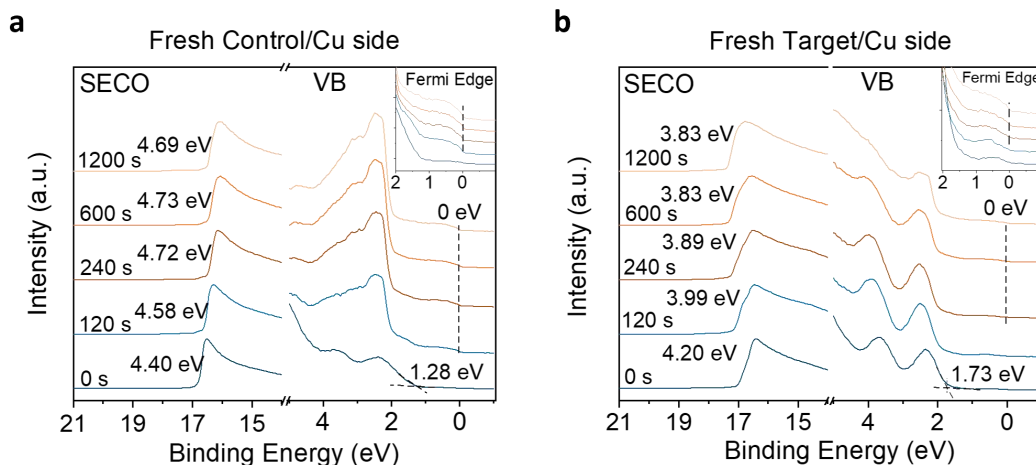
Supplementary Fig. 2. Sample architectures of the Cu and substrate sides from the delaminated full device using a mechanical peel-off approach.



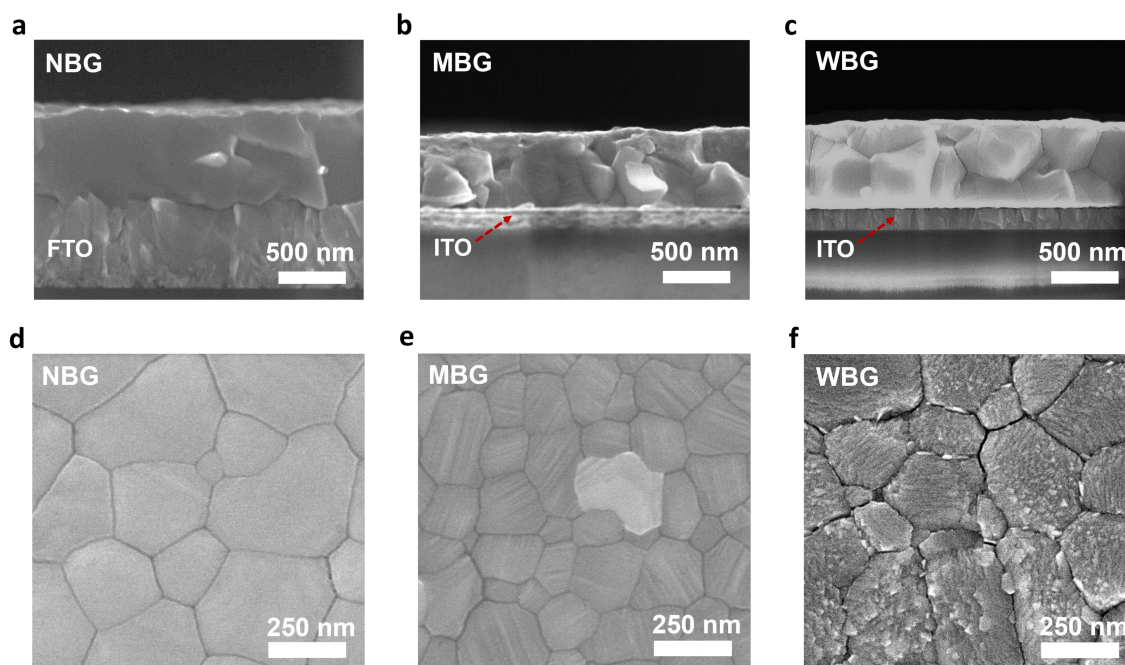
Supplementary Fig. 3. Top-view scanning electron microscopy (SEM) images of the peeled-off samples with architectures of C_{60}/Cu **(a)** and $C_{60}/YbO_x/Cu$ **(b)**. C_{60} is the outermost layer in these cases, where the Cu electrode is the underlayer.



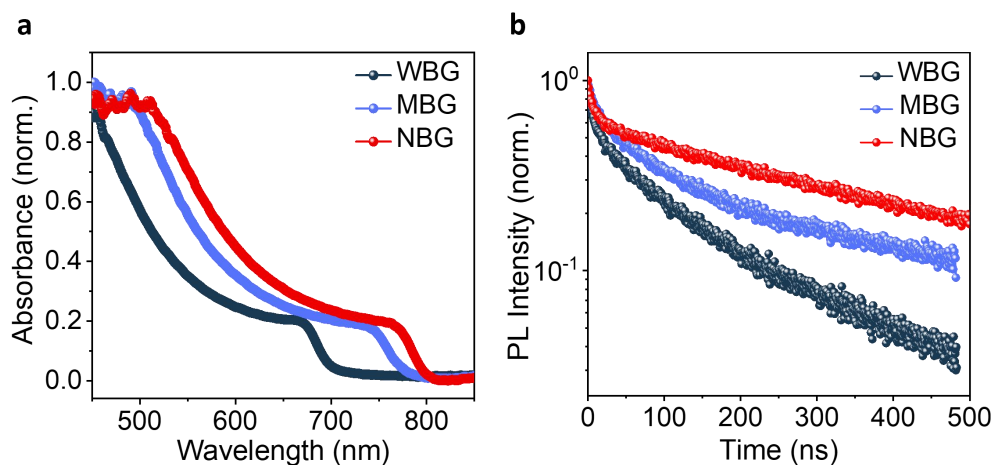
Supplementary Fig. 4. **a**, Photograph of the ageing conditions of the unencapsulated p-i-n PSCs with and without the YbO_x buffer layers. The PSCs devices were heated at about 80°C in ambient air [relative humidity (RH) of 80%]. **b-c**, Photograph of the devices with and without YbO_x after ageing for 5 h.



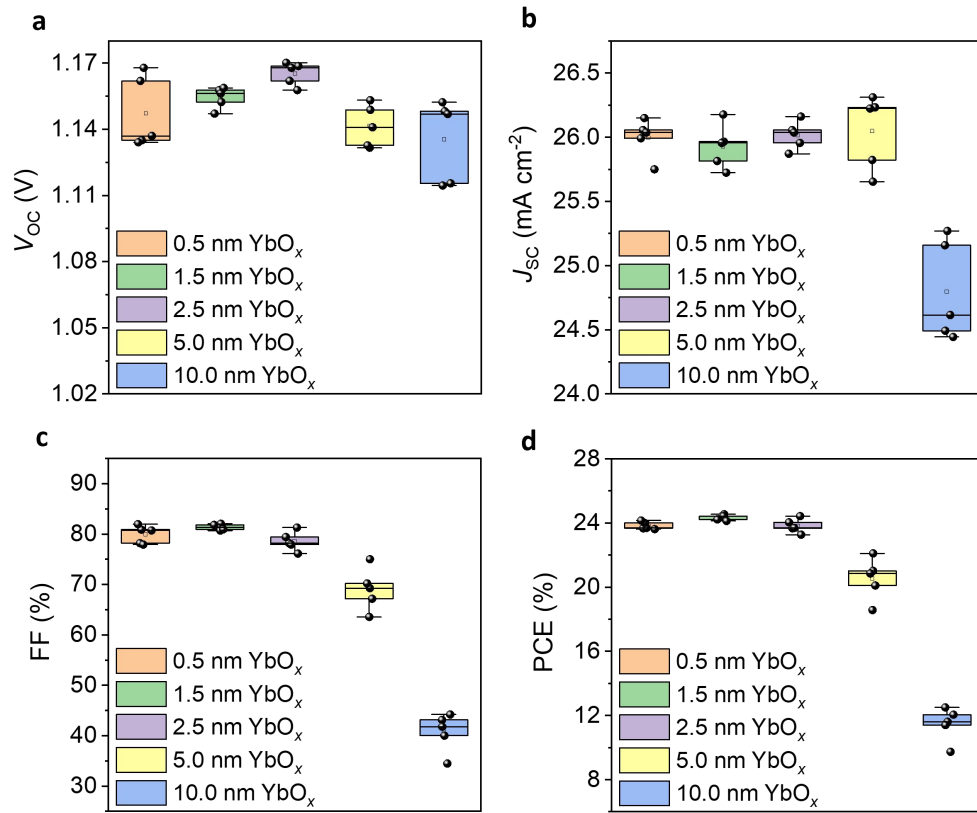
Supplementary Fig. 5. UPS depth profiling was measured at the Cu side of **(a)** the fresh control sample and **(b)** the fresh target sample. Samples architecture: Control: substrate/perovskite/ C_{60} /Cu; Target: substrate/perovskite/ C_{60} /YbO_x/Cu. Note that the composition of the perovskite film is $FA_{0.83}Cs_{0.17}Pb(I_{0.6}Br_{0.4})_3$. The occurrence of electronic metal states near the Fermi level edge (1200 s) provides a good indicator of removing the organic C_{60} and C_{60} /YbO_x attached to the Cu electrode.



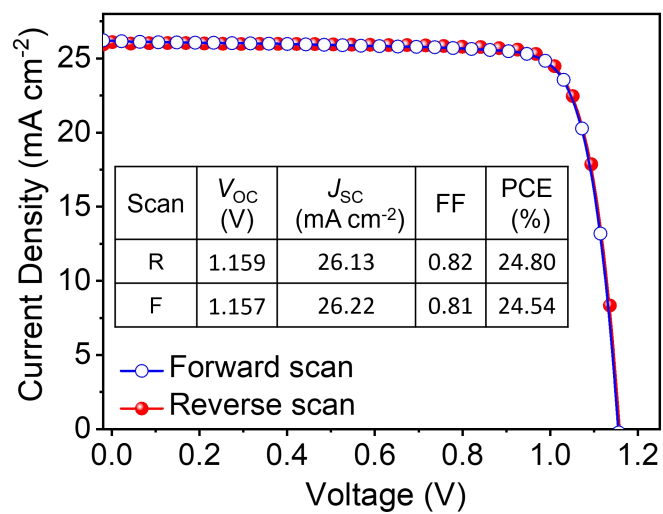
Supplementary Fig. 6. **a-c**, Cross-sectional SEM images of narrow-bandgap (NBG), mid-bandgap (MBG) and wide-bandgap (WBG) perovskites deposited on the substrates used in the real PSCs. NBG: the glass/FTO/MeO-2PACz substrate. MBG: glass/ITO/PTAA substrate. WBG: glass/ITO/NiO_x/MeO-2PACz&2PACz substrate. **d-f**, The corresponding top-view SEM images.



Supplementary Fig. 7. a, Ultraviolet-visible (UV-vis) absorption spectra of the WBG, MBG and NBG perovskites. **b**, Time-resolved photoluminescence (TRPL) decays of the NBG, MBG and WBG perovskites deposited on the substrates used in PSCs. NBG samples: glass/FTO/MeO-2PACz/perovskite; MBG samples: glass/ITO/PTAA/perovskite; WBG samples: glass/ITO/NiO_x/MeO-2PACz&2PACz/perovskite.



Supplementary Fig. 8. Device performance of the NBG-PSCs with varied thickness of the YbO_x buffer layer under simulated AM 1.5G solar irradiation at 100 mW cm^{-2} . The devices were measured from the reverse scan (RS, from 1.22 V to -0.02 V).



Supplementary Fig. 9. J – V characteristic of the best-performing NBG-PSCs obtained from reverse (R) and forward (F) scans under AM 1.5G illumination at 100 mW cm^{-2} .

Insert: the photovoltaic performance parameters.

TEST REPORT

Report No: PWQC-WT-P23060221-1R

Sample Name : Photovoltaic cell

Client : Prof. Rui Zhu's Group, Peking University

No.5 Yinyuan Road, Haidian District,
Client Address : Beijing, P.R.China

Type of Project : Consignation

PHOTOVOLTAIC AND WIND POWER SYSTEMS QUALITY TEST CENTER, IEE,
CHINESE ACADEMY OF SCIENCES

June, 05, 2023

PHOTOVOLTAIC AND WIND POWER SYSTEMS QUALITY TEST CENTER, IEE,
CHINESE ACADEMY OF SCIENCES

Report No: PWQC-WT-P23060221-1R

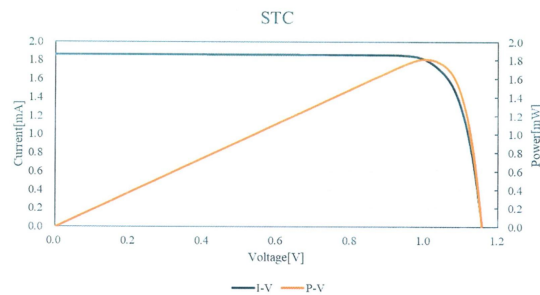
Sample code	DC2023a043
Sample S/N	1
Type	Single Junction Inverted Perovskite Solar Cell
Designated area	0.07221cm ² The designated area was offered by National Institute of Metrology, China. Test Report No. CDJ2023-02144.

Items of testing	Measurement of photovoltaic current-voltage characteristics					
Sample code	DC2022a043					
Results	Voltage Sweep	Isc (mA)	Jsc (mA/cm ²)	Voc (V)	Pm (mW)	File
		1.864	25.813	1.155	1.811	A20230602 52747
	Reverse	Ipm (mA)	Vpm (V)	FF (%)	Er (%)	
		1.811	1.000	84.17	25.09	
	Voltage Sweep	Isc (mA)	Jsc (mA/cm ²)	Voc (V)	Pm (mW)	File
		1.870	25.888	1.150	1.773	A20230602 52825
	Forward	Ipm (mA)	Vpm (V)	FF (%)	Er (%)	
		1.773	1.000	82.48	24.55	
Measurement uncertainty:						
Use(Isc)=1.9% (k=2)						
Use(Voc)=1.8% (k=2)						
Use(Pm)=2.5% (k=2)						

Page 2 of 4

PHOTOVOLTAIC AND WIND POWER SYSTEMS QUALITY TEST CENTER, IEE, CHINESE ACADEMY OF SCIENCES

Report No: PWQC-WT-P23060221-1R

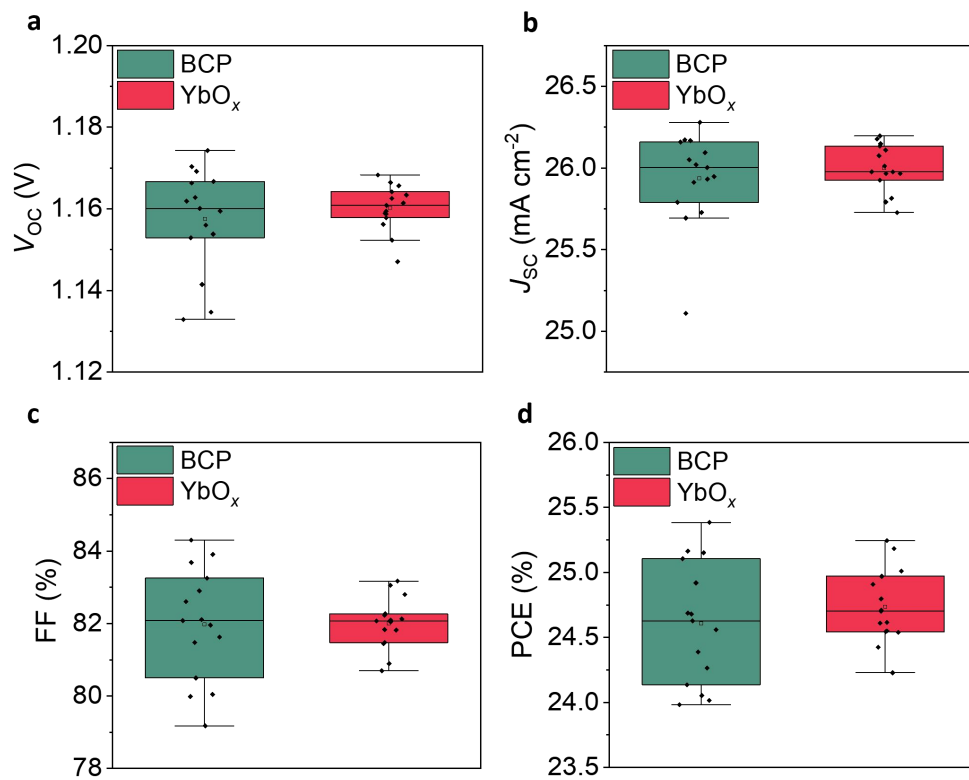


Type Single Junction Inverted Perovskite Solar Cell
Ser.No DC2023a043
Area 0.07221 cm²
Isc 1.864 mA
Jsc 25.813 mA/cm²
Voc 1.155 V
FF 84.17 %
Pm 1.811 mW
Er 25.09 %
Ipm 1.811 mA
Vpm 1.000 V
Voltage Sweep Reverse
Sweep time 1.89 s
Temp 25 °C
Irr 100 mW/cm²
File A20230602152747

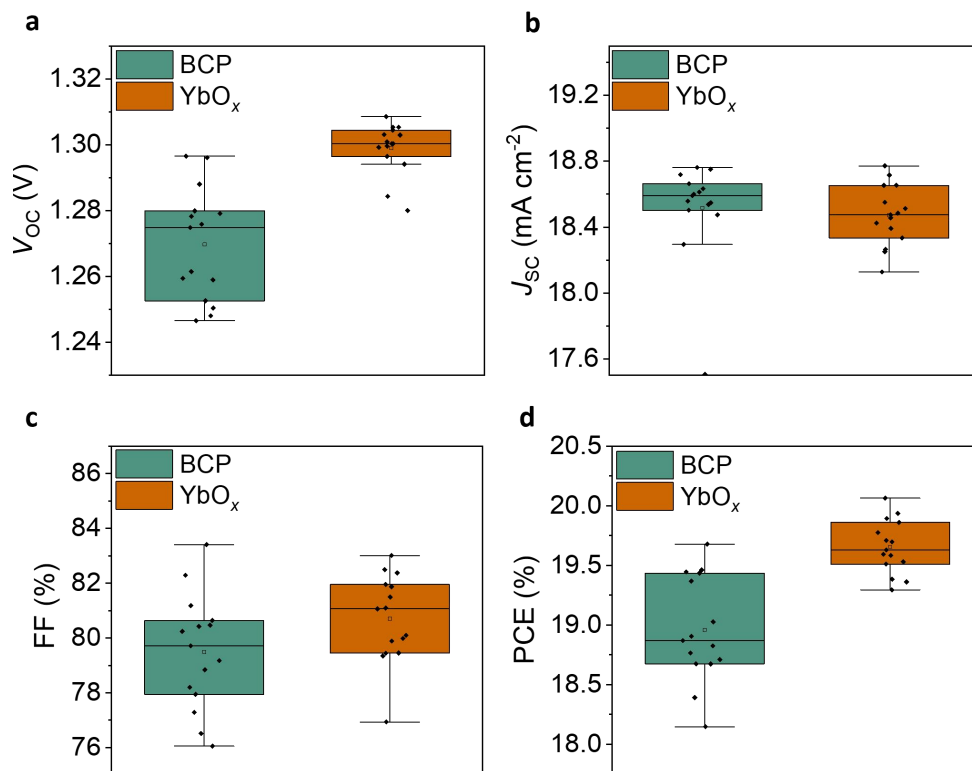
Page 3 of 4



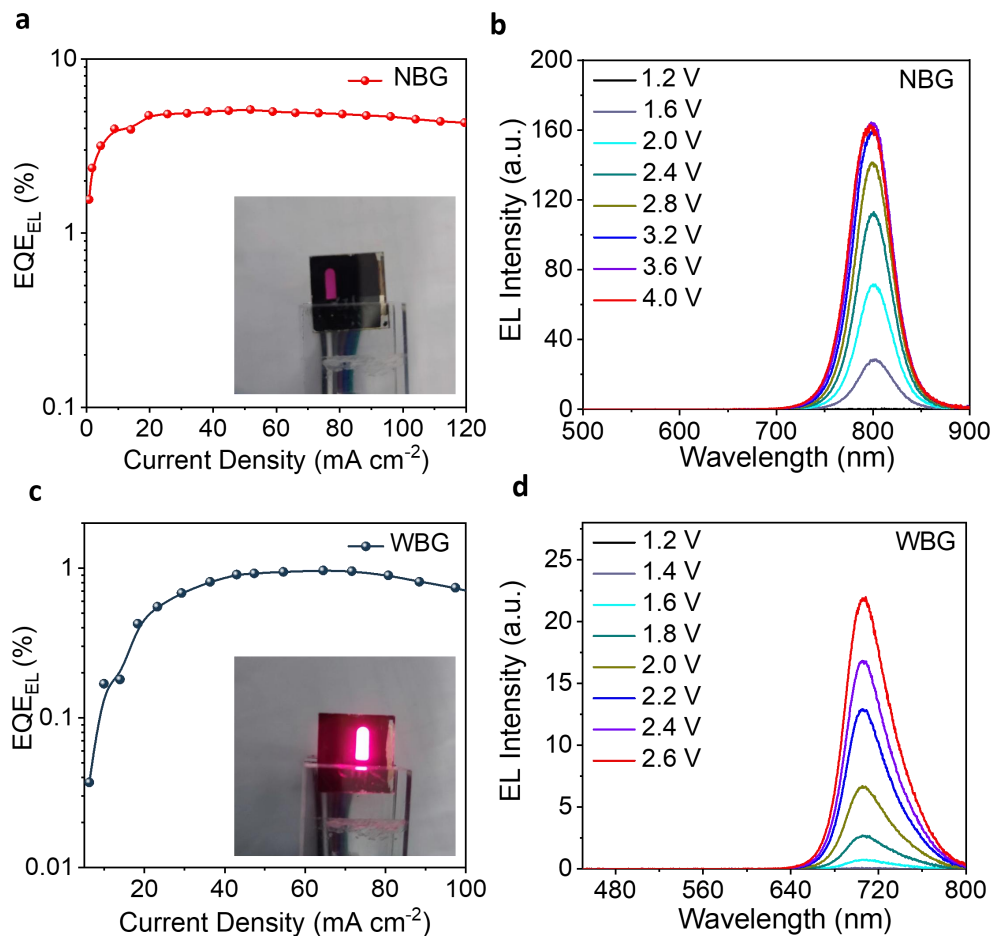
Supplementary Fig. 10. Independent PCE testing report of a p-i-n PSC with YbO_x by an accredited laboratory of Photovoltaic and Wind Power Systems Quality Test Center (Chinese Academy of Science, China), showing a certified PCE of 25.09% (a J_{SC} of 25.81 mA cm⁻², a V_{OC} of 1.155 V, and an FF of 84.17%).



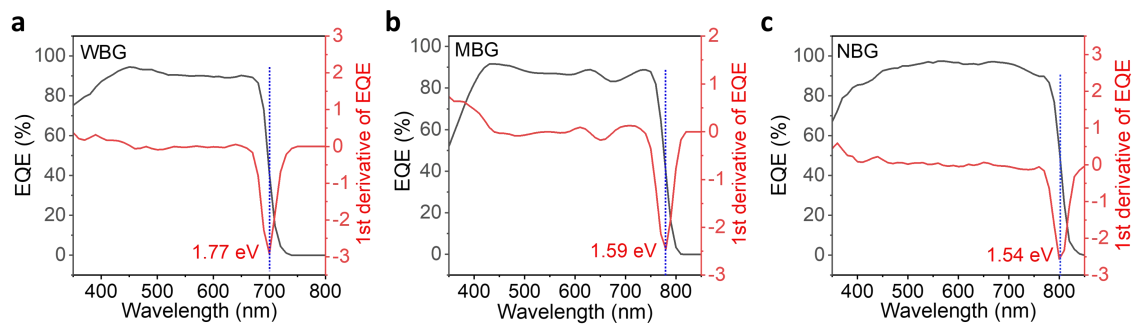
Supplementary Fig. 11. Comparison of V_{oc} (a), J_{sc} (b), FF (c), PCE parameters (d) of the NBG-containing PSCs with BCP or YbO_x buffer layers.



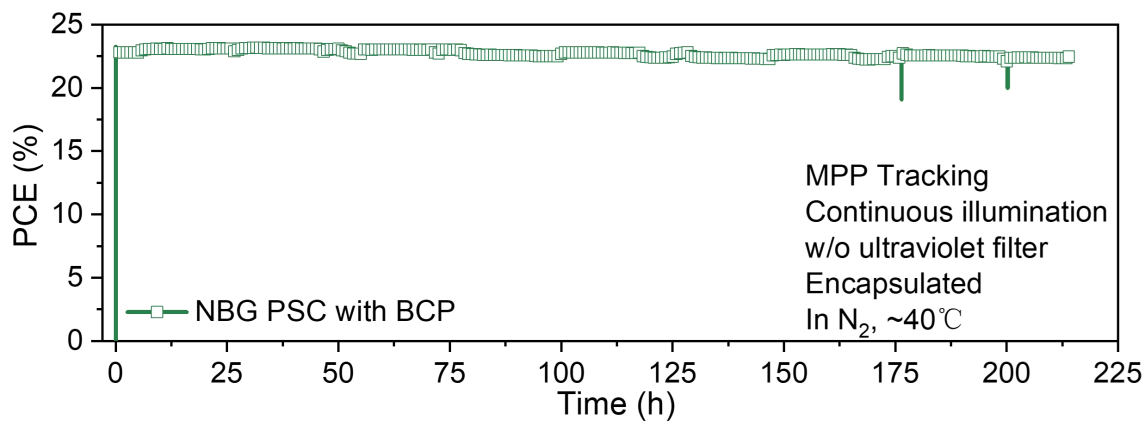
Supplementary Fig. 12. Comparison of V_{OC} (a), J_{SC} (b), FF (c), PCE parameters (d) of the WBG-PSCs with BCP or YbO_x buffer layers.



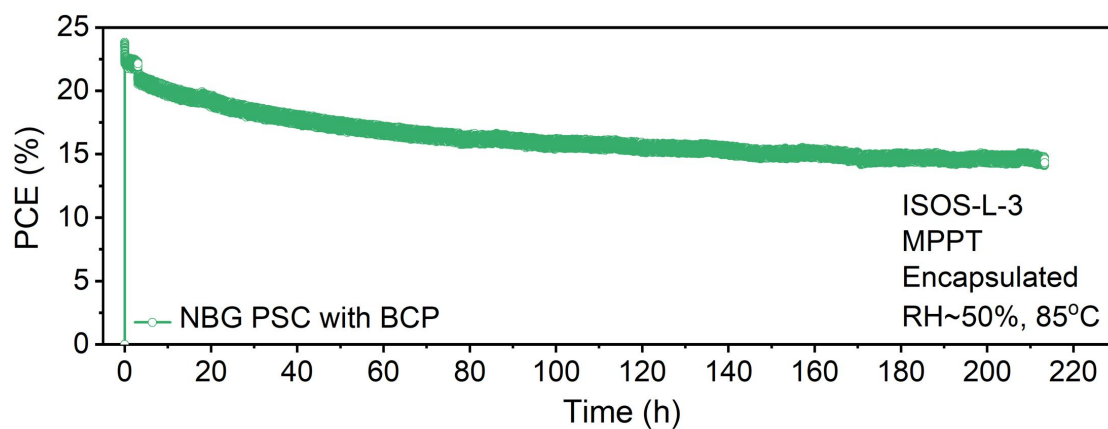
Supplementary Fig. 13. The external quantum efficiency of the electroluminescence (EQE_{EL}) as a function of the injected current density for the NBG- (**a**) and WBG-containing (**c**) devices. Insets: photographs of the NBG or WBG devices, respectively, under a bias ~ 2.5 V. Room-temperature electroluminescence (EL) spectra of the NBG- (**b**) and WBG-containing (**d**) PSCs under the different bias operating as a light-emitting diode (LED).



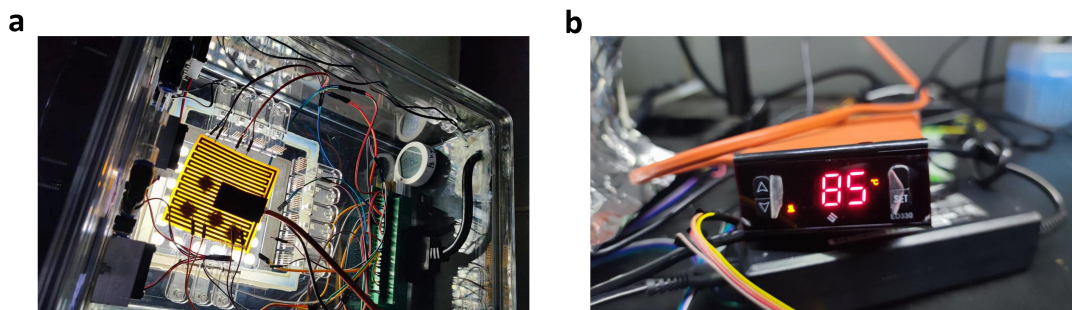
Supplementary Fig. 14. Bandgaps of WBG (a), MBG (b) and NBG (c) perovskite absorbers determined from the derivative of corresponding EQE spectra^{9,10}.



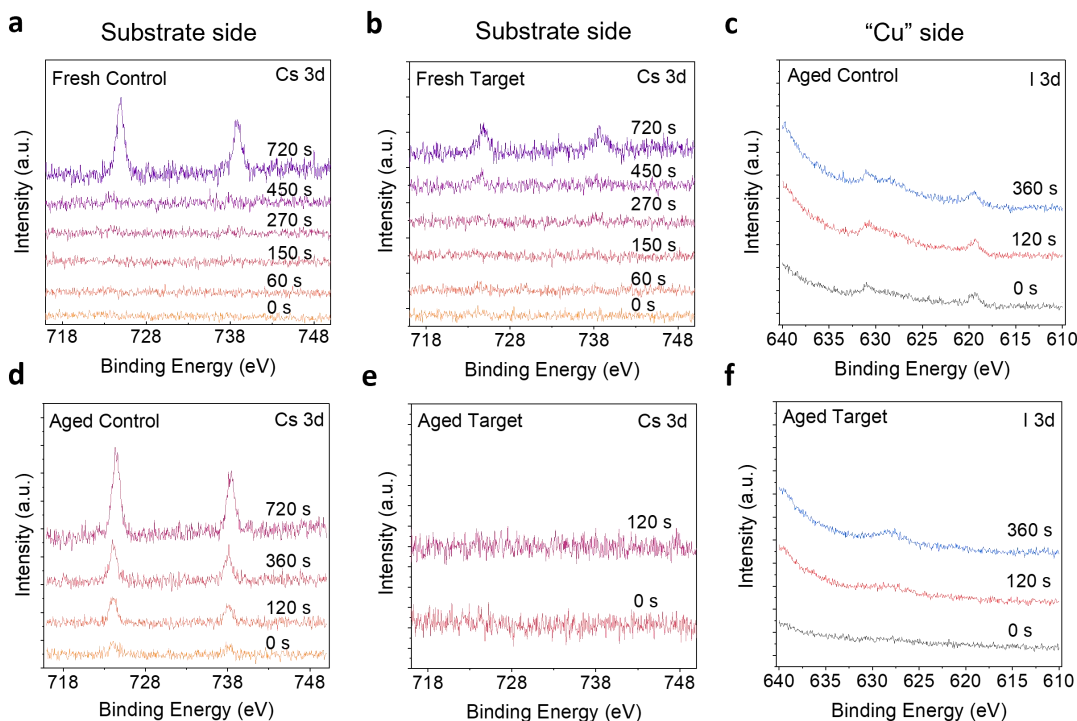
Supplementary Fig. 15. Continuous MPPT of an encapsulated NBG-PSCs with BCP kept at about 40°C under continuous white LED arrays (with no ultraviolet filters) with full-spectrum white LED illumination (continuous 1-sun illumination) in a nitrogen atmosphere. After 213 h, the device retained 96% of its initial PCE.



Supplementary Fig. 16. The MPPT curve of an encapsulated NBG-PSC with BCP held at 85°C under full-spectrum white LED continuous illumination (continuous 1-sun illumination) and 50% relative humidity (ISOS-L-3 conditions).



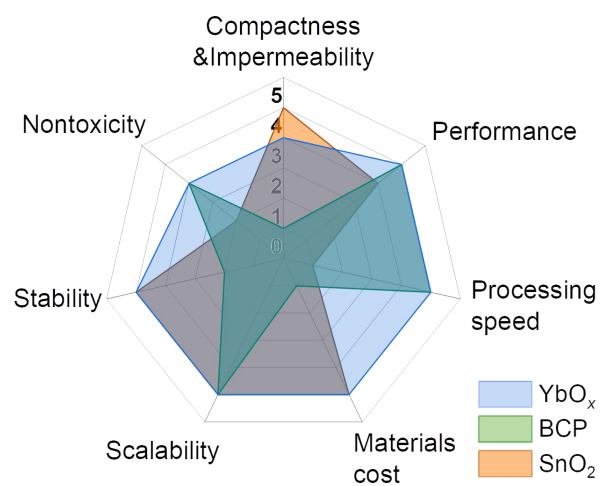
Supplementary Fig. 17. Photographs of the MPPT operations under the ISOS-L-3 protocol in our lab. The ageing devices adhered to the polyimide heating film (**a**), and a digital-displayed temperature controller controlled and monitored the temperature (**b**).



Supplementary Fig. 18. XPS depth profiling using gas cluster ion beam (GCIB) sputtering. Cs 3d XPS spectra: the fresh control (without YbO_x) sample (a) and aged control sample (d) after thermal annealing at 80°C in ambient air for 5 h detected at the substrate side. The fresh target (with YbO_x) sample (b) and aged target sample (e) after thermal annealing at 80°C in ambient air for 5 h detected at the substrate side. I 3d XPS spectra: the aged control (c) and target (f) samples after thermal annealing at 80°C in ambient air for 5 h detected at the Cu side.

After thermal ageing, in the aged control sample without YbO_x , Cs 3d XPS peaks are detectable at the substrate side of the peeled-off samples, indicating the outward migration of Cs^+ . In contrast, such characteristic peaks are undetectable for the aged target samples with YbO_x . Additionally, in the aged target samples, I 3d XPS peaks (at $\sim$

619 eV) are unfound at the Cu side of the peeled-off samples. In contrast, these characteristic peaks are evident throughout the sputtering process, indicating the severe diffusion of I^- from the perovskite layer.



Supplementary Fig. 19. Qualitative summary of seven essential properties of YbO_x, BCP, and SnO₂ based on related literature¹¹⁻¹⁵.

Supplementary Table 1. The commercialization strengths of the YbO_x buffer compared to BCP and SnO₂ materials.

	YbO _x	SnO ₂	BCP
Cost			
Film thickness per device (nm)	1.5	20	8
Usage for 1 m ² solar panel (~200 W) (m ³)	15×10 ⁻⁹	22×10 ⁻⁹	80×10 ⁻⁹
Usage for 1000 GW (the total cumulative installed capacity by 2022 ~1185 GW) (m ³)	75	111	400
Total mass use for 1000 GW (tonne)	525	811	480
Raw materials price (USD g ⁻¹) ^a	0.36	67	124
The estimate buffer cost for 1000 GW (10 ⁶ USD)	189	54,337	59,520
Processing			
Processing duration (min) ^b	10 ⁻¹	10 ²	10 ¹
Reserves			
Global reserves (tonne) ¹⁶	1,000,000	5,000,000	/
Production per year globally (tonne) ¹⁶	50	380,400	/
Total years for consumption at a rate of 200 GW per year globally	~70,000	~35,000	/

^a Note that the unit price of the metal Yb and tetrakis(dimethylamino)tin(IV) (TDMASn) are given as they are the raw materials for producing YbO_x and SnO₂, respectively. Specifically, the metal Yb is a kind of rare-earth metal and can be mass-produced with a unit price of ~0.36 USD g⁻¹ as the quote for orders of ~1 kg¹⁷. While the TDMASn

usually used as the precursor for atomic layer deposition (ALD)-grown SnO₂ is not a mass-produced commodity with a unit price of ~80 USD g⁻¹ as the quote for lab-level orders (~5 g)¹⁸. Similarly, the BCP is neither a mass-produced commodity with a unit price of ~124 USD g⁻¹ as the quote for lab-level orders (~1 g)¹⁹, either.

^b Thermal evaporation has proven to be a scalable processing technique. In our work, the estimated deposition duration for upscaling fabrication is based on a 100 cm² substrate, whose size is comparable to commercially available silicon solar cells. Note that the deposition duration for ALD SnO₂ will decrease significantly when replacing the classic temporal ALD instrument with a high-throughput spatial ALD instrument. However, the cost for the spatial ALD instrument will be increased significantly (e.g., the quotation from the company is up to 3-fold higher than that of the current classic ALD instrument)³⁻⁵. In addition, it is reported that the PSCs based on spatial ALD SnO₂ show unpleasant performance, indicating much more research and development for the spatial ALD technique is required^{4,20}.

Supplementary Table 2. The cost of Yb compared to the common metals including Au, Ag, and Cu.

Metal	Yb ¹⁷	Au ²¹	Ag ²¹	Cu ²¹
Price (USD kg ⁻¹)	360	60,000	740	120

Supplementary Table 3. Details of the oxidization process of Yb⁰ surfaces exposed in the XPS chamber.

Pressure (torr)	Time (s)	Exposure (L ^a)	Yb ⁰	Yb ³⁺	Yb ³⁺ :Yb ⁰	d (nm)
4.2×10 ⁻¹⁰	2.4×10 ³	1	100	0	0	0
8.4×10 ⁻⁹	2.0×10 ³	30	85.2	14.8	0.17	0.57
1.7×10 ⁻⁸	1.8×10 ³	60	80.4	19.6	0.24	0.74
1.3×10 ⁻⁸	4.5×10 ³	120	72.1	27.9	0.39	1.02
2.8×10 ⁻⁸	4.3×10 ³	240	64.1	35.9	0.56	1.3
4.2×10 ⁻⁸	5.7×10 ³	480	58.9	41.1	0.7	1.48

^aL represents Langmuir, 1 L=10⁻⁶ torr·s=1.33×10⁻⁴ Pa·s.

Supplementary Table 4. Photovoltaic performance of the NBG PSCs with and without an antireflection layer (125 nm of MgF₂) under simulated AM 1.5G solar irradiation at 100 mW cm⁻². The devices were measured from reverse scans (RS, from 1.22 V to -0.02 V).

Sample	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF	PCE (%)
w/o MgF ₂	1.17	25.6	0.83	24.9
w/ MgF ₂	1.16	26.1	0.83	25.2

Supplementary Table 5. Summary of devices parameters and champion PCEs of state-of-the-art p-i-n PSCs with an NBG (1.53–1.56 eV) from the literature.

Perovskite composition	E_g (eV)	Cathode interlayer	Deposition method	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF	PCE (%)	Ref.
CS _{0.05} (FA _{0.95} MA _{0.05}) _{0.95} Pb(I _{0.95} Br _{0.05}) ₃	1.56	BCP	Spin coating	1.20	25.2	0.82	24.7	22
-	1.56	BCP	Spin coating	1.19	25.7	0.85	25.9 (25.4) ^a	23
CS _{0.05} (FA _{0.95} MA _{0.05}) _{0.95} Pb(I _{0.95} Br _{0.05}) ₃	1.56	BCP	Physical vapor deposition	1.20	24.8	0.85	25.2 (25.4) ^a	24
CS _{0.05} (FA _{0.95} MA _{0.05}) _{0.95} Pb(I _{0.95} Br _{0.05}) ₃	1.55	BCP	Physical vapor deposition	1.21	25.1	0.84	25.5 (24.7) ^a	25
CS _{0.05} (FA _{0.98} MA _{0.02}) _{0.95} Pb(I _{0.98} Br _{0.02}) ₃	1.55	BCP	Physical vapor deposition	1.19	24.9	0.84	24.5	26
FA _{0.95} CS _{0.05} PbI ₃	-	SnO ₂	Atomic layer deposition	1.16	25.7	0.83	24.5	27
CS _{0.05} (FA _{0.98} MA _{0.02}) _{0.95} Pb(I _{0.98} Br _{0.02}) ₃	-	BCP	Physical vapor deposition	1.17	25.5	0.83	24.9 (24.5) ^a	28
CS _{0.05} (FA _{0.98} MA _{0.02}) _{0.95} Pb(I _{0.98} Br _{0.02}) ₃	1.55	BCP	Physical vapor deposition	1.18	24.8	0.84	24.6 (24.2) ^a	9
(FA _{0.98} MA _{0.02}) _{0.95} CS _{0.05} Pb(I _{0.95} Br _{0.02}) ₃	1.55	BCP	Physical vapor deposition	1.18	25.7	0.82	25.0 (24.3) ^a	29
CS _{0.03} (FA _{0.90} MA _{0.10}) _{0.97} PbI ₃	1.55	BCP	Physical vapor deposition	1.21	24.5	0.82	24.3	30

$\text{Cs}_{0.05}(\text{MA}_{0.05}\text{FA}_{0.95})_{0.95}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3$	1.55	SnO_2	Atomic layer deposition	1.17	25.0	0.83	24.3 (24.2) ^a	31
FAMAPbI ₃	1.55	BCP	Physical vapor deposition	-	-	-	23.6	30
$\text{Cs}_{0.05}\text{FA}_{0.82}\text{MA}_{0.1}\text{GUA}_{0.03}\text{PbI}_3(\text{GUA}_{0.03})$	1.55	BCP	Physical vapor deposition	1.17	24.1	0.82	22.9	32
$\text{Rb}_{0.05}\text{Cs}_{0.05}\text{MA}_{0.05}\text{FA}_{0.85}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3$	1.54	YbO_x	Physical vapor deposition	1.16	26.1	0.83	25.2 (25.09)^a	This work
$\text{Rb}_{0.05}\text{Cs}_{0.05}\text{MA}_{0.05}\text{FA}_{0.85}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3$	1.53	BCP	Physical vapor deposition	1.15	26.1	0.85	25.5 (24.1) ^a	33
$\text{Cs}_{0.05}\text{FA}_{0.85}\text{MA}_{0.1}\text{PbI}_3$	1.53	BCP	Physical vapor deposition	1.15	24.9	0.83	23.9 ^c	34

^acertified values

Supplementary Table 6. Summary of device performance of p-i-n PSCs with inorganic or bilayer buffer layers containing an inorganic buffer layer.

Device configuration	Buffer layer	PCE (%)	Year	Ref.
FTO/NiMgLiO/MAPbI ₃ /PCBM/BCP/Bi/Ag	BCP/Bi	18.1	2018	35
FTO/NiMgLiO/MAPbI ₃ /PCBM/Ti(Nb)O _x /Ag	Ti(Nb)O _x	18.3	2015	36
ITO/NiO _x -KCl/Cs _{0.07} FA _{0.93} PbI ₃ /C ₆₀ /SnO ₂ /IZO/Ag	SnO ₂ /IZO	20.3	2022	30
Mo/IZO/PTAA/perovskite/LiF/C ₆₀ /SnO ₂ /ZTO/IZO/Ag	SnO ₂ /ZTO/IZO	21.0	2023	37
ITO/NiO _x /CsMAFAPb(IBr) ₃ /PCBM/C ₆₀ /BCP/Cr/Au	BCP/Cr	22.1	2021	38
ITO/PTAA/ Cs _{0.05} (FA _{0.92} MA _{0.08}) _{0.95} Pb(I _{0.92} Br _{0.08}) ₃ /PCBM/BCP/VO ₂ /Cu	BCP/VO ₂	23.1	2021	39
ITO/PTAA/Cs _{0.05} FA _{0.95} PbI ₃ /C ₆₀ /SnO ₂ /a-ZrN _x /Cu	SnO ₂ /a-ZrN _x	23.1	2023	40
FTO/NiO _x /Me-4PACz/FA _{0.95} Cs _{0.05} PbI ₃ /C ₆₀ /SnO ₂ /Cr/Cu	SnO ₂ /Cr	23.4	2023	27
ITO/2PACz/Cs _{0.05} (MA _{0.05} FA _{0.95}) _{0.95} Pb(I _{0.95} Br _{0.05}) ₃ /C ₆₀ /SnO ₂ /Ag	SnO ₂	24.2	2022	31
FTO/NiO _x /Me-4PACz/FA _{0.95} Cs _{0.05} PbI ₃ /C ₆₀ /SnO ₂ /Ag	SnO ₂	24.5	2023	27

**FTO/MeO-2PACz/Rb_{0.05}Cs_{0.05}MA_{0.05}FA_{0.85} Pb(I_{0.95}Br_{0.05})₃
/PCBM/YbO_x/Cu**

YbO_x

25.09 (certified)

2023

**This
work**

Supplementary Table 7. Summary of devices parameters and champion PCEs of state-of-the-art p-i-n PSCs with a WBG (1.75–1.80 eV) from the literature.

Perovskite composition	E_g (eV)	Buffer layer	Deposition method	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF	PCE (%)	Ref.
DMA _{0.1} CS _{0.4} Br _{0.25} Cl _{0.05}	1.79	BCP	Physical vapor deposition	1.26	17.4	0.80	17.7	41
FA _{0.8} CS _{0.2} Pb(I _{0.6} Br _{0.4}) ₃	1.79	SnO ₂	Atomic layer deposition	1.33	17.3	0.84	19.3	42
FA _{0.75} CS _{0.25} Pb(I _{0.6} Br _{0.4}) ₃	1.79	BCP	Physical vapor deposition	1.26	17.9	0.79	17.8	43
FA_{0.83}CS_{0.17}Pb(I_{0.6}Br_{0.4})₃	1.77	YbO_x	Physical vapor deposition	1.30	18.6	0.83	20.1	This work
(FA _{0.6} MA _{0.4}) _{0.9} CS _{0.1} Pb(I _{0.6} Br _{0.4}) ₃	1.77	BCP	Physical vapor deposition	1.26	18.1	0.80	18.3	44
FA _{0.83} CS _{0.17} Pb(I _{0.6} Br _{0.4}) ₃	1.77	BCP	Physical vapor deposition	1.22	-	≈ 0.80	≈ 17	45
FA _{0.8} CS _{0.2} Pb(I _{0.6} Br _{0.4}) ₃	1.77	BCP	Physical vapor deposition	1.28	17.2	0.80	17.7	46
FA _{0.8} CS _{0.2} Pb(I _{0.6} Br _{0.4}) ₃	1.77	BCP	Physical vapor deposition	1.22	17.0	0.80	16.5	47
FA _{0.8} CS _{0.2} Pb(I _{0.62} Br _{0.38}) ₃	1.77	BCP	Physical vapor deposition	1.22	17.4	0.82	17.3	48
FA _{0.8} CS _{0.2} Pb(I _{0.6} Br _{0.4}) ₃	1.77	SnO ₂	Atomic layer deposition	1.21	17.1	0.77	15.9	49

$\text{Cs}_{0.3}\text{FA}_{0.6}\text{DMA}_{0.1}\text{Pb}(\text{I}_{0.7}\text{Br}_{0.3})_3$	1.75	BCP	Physical vapor deposition	1.32	18.7	0.82	20.3	50
$\text{FA}_{0.7}\text{Cs}_{0.3}\text{Pb}(\text{I}_{0.7}\text{Br}_{0.3})_3$	1.75	SnO_2	Atomic layer deposition	1.23	17.5	0.80	17.1	51
$\text{FA}_{0.7}\text{Cs}_{0.3}\text{Pb}(\text{I}_{0.7}\text{Br}_{0.3})_3$	1.75	BCP	Physical vapor deposition	1.14	16.4	0.75	14.0	52

Supplementary Table 8. Summary of enhanced device stability of p-i-n PSCs with an effective buffer layer. The strength was evaluated under accelerated ageing conditions according to International Summit on Organic Photovoltaic Stability (ISOS) protocols.

Initial PCE (%)	Buffer layer	Multiple stresses	Duration	Relative PCE retained (%)	Ref.
23.05	SnO ₂	MPP at 65°C in ambient air (50% RH)	500 h	~ 92	34
~20	SnO ₂ /Cr	MPP at 85°C in N ₂	1500 h	~90	27
23.6	YbO_x	MPP at 85°C in ambient air (50% RH)	500 h	~85	This work

References

- 1 Lide, D. R. *CRC handbook of chemistry and physics*. Vol. 85 (CRC press, 2004).
- 2 Man, J.-X., Hu, J.-T., Wang, D.-K., He, S.-J. & Lu, Z.-H. Ytterbium oxide electron injection interface in organic light-emitting diode. *Appl. Phys. Lett.* **120**, 121101 (2022).
- 3 Chang, N. L. *et al.* Techno-economic analysis of the use of atomic layer deposited transition metal oxides in silicon heterojunction solar cells. *Prog. Photovolt. Res. Appl.* **31**, 414-428 (2022).
- 4 Poodt, P. *et al.* Spatial atomic layer deposition: A route towards further industrialization of atomic layer deposition. *J. Vac. Sci. Technol. A* **30**, 010802 (2012).
- 5 Chen, M. *et al.* Atmospheric-pressure atomic layer deposition: Recent applications and new emerging applications in high-porosity/3D materials. *Dalton Trans.* **52**, 10254-10277 (2023).
- 6 Mott, N. Electrons in glass. *Nature* **257**, 15-18 (1975).
- 7 Rau, U. Reciprocity relation between photovoltaic quantum efficiency and electroluminescent emission of solar cells. *Phys. Rev. B* **76**, 085303 (2007).
- 8 Shockley, W. & Queisser, H. J. Detailed balance limit of efficiency of p-n junction solar cells. *J. Appl. Phys.* **32**, 510-519 (1961).
- 9 Li, G. *et al.* Highly efficient p-i-n perovskite solar cells that endure temperature variations. *Science* **379**, 399-403 (2023).

- [KCQjwz8emBhDrARIsANNJJS7c61MGFMyxwrRcpA4_BM3CO3OVf6GIAhgd6IYWwtA0zITUXH8VQasaAneXEALw_wcB&gclid=aw.ds](#) (2023).
- 20 Hoffmann, L. *et al.* Spatial atmospheric pressure atomic layer deposition of tin oxide as an impermeable electron extraction layer for perovskite solar cells with enhanced thermal stability. *ACS Appl. Mater. Interfaces* **10**, 6006-6013 (2018).
 - 21 Wall Street Journal. Markets data. *WSJ* https://www.wsj.com/market-data?mod=nav_top_subsection (2023).
 - 22 Huang, Y. C. *et al.* Finite perovskite hierarchical structures via ligand confinement leading to efficient inverted perovskite solar cells. *Energy Environ. Sci.* **16**, 557-564 (2023).
 - 23 Tan, Q. *et al.* Inverted perovskite solar cells using dimethylacridine-based dopants. *Nature* **620**, 545-551 (2023).
 - 24 Zhang, S. *et al.* Minimizing buried interfacial defects for efficient inverted perovskite solar cells. *Science* **380**, 404-409 (2023).
 - 25 Peng, W. *et al.* Reducing nonradiative recombination in perovskite solar cells with a porous insulator contact. *Science* **379**, 683-690 (2023).
 - 26 Zheng, X. P. *et al.* Co-deposition of hole-selective contact and absorber for improving the processability of perovskite solar cells. *Nat. Energy* **8**, 462 (2023).
 - 27 Li, C. *et al.* Rational design of lewis base molecules for stable and efficient inverted perovskite solar cells. *Science* **379**, 690-694 (2023).

- 28 Wu, X. *et al.* Backbone engineering enables highly efficient polymer hole-transporting materials for inverted perovskite solar cells. *Adv. Mater.* **35**, 2208431 (2023).
- 29 Li, Z. *et al.* Organometallic-functionalized interfaces for highly efficient inverted perovskite solar cells. *Science* **376**, 416-420 (2022).
- 30 Azmi, R. *et al.* Damp heat-stable perovskite solar cells with tailored-dimensionality 2D/3D heterojunctions. *Science* **376**, 73-77 (2022).
- 31 Zhu, Z. *et al.* Correlating the perovskite/polymer multi-mode reactions with deep-level traps in perovskite solar cells. *Joule* **6**, 2849-2868 (2022).
- 32 Chen, B. *et al.* Passivation of the buried interface via preferential crystallization of 2D perovskite on metal oxide transport layers. *Adv. Mater.* **33**, 2103394 (2021).
- 33 Jiang, Q. *et al.* Surface reaction for efficient and stable inverted perovskite solar cells. *Nature* **611**, 278-283 (2022).
- 34 Chen, H. *et al.* Quantum-size-tuned heterostructures enable efficient and stable inverted perovskite solar cells. *Nat. Photonics* **16**, 352 (2022).
- 35 Wu, S. *et al.* A chemically inert bismuth interlayer enhances long-term stability of inverted perovskite solar cells. *Nat. Commun.* **10**, 1-10 (2019).
- 36 Chen, W. *et al.* Efficient and stable large-area perovskite solar cells with inorganic charge extraction layers. *Science* **350**, 944-948 (2015).

- 37 Li, T. *et al.* Understanding the interfacial reactions and band alignment for efficient and stable perovskite solar cells built on metal substrates with reduced upscaling losses. *Adv. Mater.* **35**, 2211959 (2023).
- 38 Cao, Q. *et al.* Efficient and stable inverted perovskite solar cells with very high fill factors via incorporation of star-shaped polymer. *Sci. Adv.* **7**, eabg0633 (2021).
- 39 Li, X. *et al.* High-efficiency and durable inverted perovskite solar cells with thermally-induced phase-change electron extraction layer. *Adv. Energy Mater.* **11**, 2102844 (2021).
- 40 Xiao, M. *et al.* Engineering amorphous-crystallized interface of ZrNx barriers for stable perovskite solar cells. *Adv. Mater.* **35**, 2301684 (2023).
- 41 Wen, J. *et al.* Steric engineering enables efficient and photostable wide-bandgap perovskites for all-perovskite tandem solar cells. *Adv. Mater.* **34**, 2110356 (2022).
- 42 Chen, H. *et al.* Regulating surface potential maximizes voltage in all-perovskite tandems. *Nature* **613**, 676-681 (2022).
- 43 Chen, W. *et al.* Monolithic perovskite/organic tandem solar cells with 23.6% efficiency enabled by reduced voltage losses and optimized interconnecting layer. *Nat. Energy* **7**, 229-237 (2022).
- 44 Li, Z. *et al.* Minimized surface deficiency on wide-bandgap perovskite for efficient indoor photovoltaics. *Nano Energy* **78**, 105377 (2020).

- 45 Oliver, R. D. J. *et al.* Understanding and suppressing non-radiative losses in methylammonium-free wide-bandgap perovskite solar cells. *Energy Environ. Sci.* **15**, 714-726 (2022).
- 46 He, R. *et al.* Pure 2D perovskite formation by interfacial engineering yields a high open-circuit voltage beyond 1.28 V for 1.77-eV wide-bandgap perovskite solar cells. *Adv. Sci.* **9**, 2203210 (2022).
- 47 Lin, R. X. *et al.* Monolithic all-perovskite tandem solar cells with 24.8% efficiency exploiting comproportionation to suppress Sn(II) oxidation in precursor ink. *Nat. Energy* **4**, 864-873 (2019).
- 48 Lin, R. *et al.* All-perovskite tandem solar cells with improved grain surface passivation. *Nature* **603**, 73-78 (2022).
- 49 Xiao, K. *et al.* All-perovskite tandem solar cells with 24.2% certified efficiency and area over 1 cm² using surface-anchoring zwitterionic antioxidant. *Nat. Energy* **5**, 870-880 (2020).
- 50 Jiang, Q. *et al.* Compositional texture engineering for highly stable wide-bandgap perovskite solar cells. *Science* **378**, 1295-1300 (2022).
- 51 Tong, J. *et al.* Carrier control in Sn-Pb perovskites via 2D cation engineering for all-perovskite tandem solar cells with improved efficiency and stability. *Nat. Energy* **7**, 642-651 (2022).

- 52 Dewei, Z. *et al.* Efficient two-terminal all-perovskite tandem solar cells enabled by high-quality low-bandgap absorber layers. *Nat. Energy* **3**, 1093-1100 (2018).